

Reading the whole corpus: a reproducible cross-sample Raman workflow with a Spectral 3D Gaussian Splatting representation for hyperspectral maps

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Abstract

Raman mapping produces a hyperspectral cube, and turning that cube into numbers depends strongly on a chain of preprocessing choices. Two further problems appear once a laboratory has produced many maps rather than one. The corpus is analysed one file at a time, and the corpus that can be analysed at all is only the subset somebody remembered to export by hand from the instrument’s proprietary format. We present RAMANSP, a compact open framework in the spirit of RamanSPy, providing typed spectral containers, composable preprocessing protocols and an analysis layer, and extend it in three directions. First, a direct reader for the undocumented Horiba LabSpec6 container, reverse engineered here and verified bit identical against the instrument’s own text export, which removes the manual export step from the pipeline entirely. Second, a cross-sample knowledge graph that links every acquisition in a study by spectral similarity, shared unmixing endmembers and disorder-metric proximity, with each metric edge carrying the preprocessing protocol under which it was computed. Third, and as the main methodological contribution, *Spectral 3D Gaussian Splatting* (S3DGS): a map $V(x, y, \nu)$ is represented by a differentiable field of anisotropic three-dimensional ellipsoids, each localising a chemical domain in the two spatial axes and a band along the wavenumber axis, fitted by first-order descent on closed-form gradients with prune and clone densification, entirely in NumPy on a CPU. On an anonymised corpus of 108 acquisitions from a carbon mapping study, including 23 hyperspectral maps of which 57 were read straight from the vendor binary, the graph recovers 23 communities at modularity $Q = 0.68$ and automatically flags 5 groups of byte-identical duplicate exports. Across 6 maps and acquisition recipes S3DGS reconstructs held-out wavenumber channels to within 0.70 to 1.43 of each map’s own measurement noise at 8.1 to 38.0 $\times$ compression, that is, essentially to the noise floor, discarding noise rather than encoding it. We report the failure modes we hit, including one that lowered the loss while quietly corrupting the field, and specify the port to a tiled CUDA rasteriser for scaling to millions of primitives.

Keywords: Raman microscopy; hyperspectral imaging; preprocessing reproducibility; knowledge graph; Gaussian splatting; differentiable rendering; data compression; carbonaceous materials.

1 Introduction

Spontaneous Raman mapping raster scans a focused laser over a specimen and records a full spectrum at every position, producing a cube $V(x, y, \nu)$ whose third axis is Raman shift. For carbonaceous materials the cube is dominated by the D and G bands near 1350 and 1585 cm^{-1} , whose relative intensities and widths are read as a measure of structural disorder. Converting a cube into such numbers requires despiking, baseline correction, smoothing, normalisation and often curve fitting, and the answer depends on which chain was used. RamanSPy [1, 2] addressed the resulting reproducibility gap by standardising the pipeline as code: typed containers, named preprocessing *protocols*, and a benchmarkable analysis layer. That is the right foundation, and it is the one we build on. Three problems remain once a laboratory has produced many maps rather than one.

(i) The corpus is analysed one file at a time. A study running over months accumulates dozens of maps, line scans, single spectra, calibration runs and curve-fit exports, across several specimens and several acquisition recipes. Questions that matter to the experimenter, such as which samples share a spectral signature, whether disorder indices cluster by material, or whether two files are in fact the same measurement saved twice, require holding the whole study as one queryable object rather than as a directory of files.

(ii) The readable corpus is a biased subset of the measured one. Instrument vendors store data in undocumented binary containers. Analysis tooling therefore sees a file only if an operator opened it and triggered a text export by hand. Which files get exported is driven by what somebody wanted to look at that afternoon, so

the corpus available to analysis is selected by attention rather than by design. In the study analysed here, the readable text exports amounted to a single hyperspectral map; the binaries held 23.

(iii) **There is no compact, differentiable, generative representation of a map.** A cube is stored as millions of independent numbers. Denoising, super-resolution, component analysis and compression each re-derive structure from scratch, and none of them leaves behind an object that can be queried, resampled or edited. In computer vision the analogous problem was addressed by 3D Gaussian Splatting [3], which replaced dense volumetric fields and implicit networks with an explicit set of 10^5 to 10^6 differentiable anisotropic Gaussians and rapidly became a dominant scene representation. Hyperspectral variants followed [4, 5, 6], but they target novel-view synthesis from calibrated multi-view cameras. A confocal microscopy cube has no views to synthesise: it is one volume, sampled once, on a regular grid, with a physically meaningful third axis. The representation transfers; the task does not.

Contributions.

1. **RAMANSP**, a lean RamanSPy-style framework (containers, preprocessing protocols, decomposition, clustering, unmixing, spatial statistics, plotting) with a strict anonymisation layer and an automated leak gate (§4).
2. A **direct reader for the proprietary Horiba LabSpec 6 container**, verified bit identical against the vendor’s own export, which removes the manual per-file export step that otherwise decides what a corpus contains (§5).
3. A **cross-sample knowledge graph** whose content edges are spectral-similarity, shared-endmember and disorder-proximity relations, each carrying the preprocessing protocol it was computed under, and built with a mutual k -nearest-neighbour rule rather than an absolute similarity threshold for reasons specific to carbon corpora (§6).
4. **Spectral 3D Gaussian Splatting**, a differentiable field of anisotropic 3D ellipsoids fitted to a hyperspectral cube, with closed-form gradients, a separable evaluation scheme, densification, and interpretable primitives (§7).
5. An evaluation on 108 real acquisitions, including a noise-floor analysis showing that reconstruction error is limited by the acquisitions rather than by the representation (§8), and a candid account of the failure modes encountered (§9).

2 Background: Raman spectra and Raman maps

2.1 The measurement

Raman scattering is the inelastic scattering of monochromatic light by molecular and lattice vibrations, and the frequency shift between incident and scattered photons is a fingerprint of chemical bonding [7, 8]. Because the effect requires no labelling, needs little sample preparation, and is only weakly perturbed by water, it has become a general-purpose analytical probe across an unusually broad range of fields: cell and tissue biology [9, 10, 11, 12], biomedical diagnostics and oncology [13, 14, 15, 16, 17, 18], pharmaceutical analysis [19, 20], materials science and nanomaterials [21, 22], environmental monitoring [23, 24], food safety [25], mineralogy [26, 27] and forensic science [28], as well as routine analytical characterisation [29]. Surface enhancement extends it to trace concentrations [30, 31].

The weakness of the effect is the source of the analytical difficulty. The Raman cross-section is small, so spectra arrive with limited signal to noise, superimposed on a fluorescence background that is frequently orders of magnitude larger than the bands of interest, and contaminated by cosmic-ray spikes on the detector. Every quantitative Raman workflow therefore begins with a preprocessing chain.

2.2 Preprocessing as the dominant source of variance

A standard chain is cosmic-ray removal [32], baseline correction, smoothing [33] and normalisation, followed by whatever chemometric or machine-learning model is being fitted [34, 35, 36]. Baseline correction is the step most responsible for irreproducibility, and it has generated a correspondingly large literature: asymmetric least squares [37] and its adaptively reweighted variant [38], polynomial and iterative-polynomial families, and the comparative studies that try to choose between them [39, 40, 41].

The important point for this paper is that these choices are not neutral. Different defensible baselines applied to the same spectra yield materially different band ratios, and reviews of the field have repeatedly identified the absence of shared, declared preprocessing as an obstacle to comparing results between laboratories [42, 43, 44]. Large interlaboratory studies confirm the concern empirically [31]. A number reported without its protocol is therefore not fully specified, which is the principle we enforce structurally in §6: every metric edge in the knowledge graph carries the protocol under which it was computed, and edges computed under different protocols are never compared.

2.3 From spectra to hyperspectral maps

Mapping raster scans the focus and records a spectrum per position, producing a cube whose spatial axes carry chemical contrast [45, 46]. This multiplies both the promise and the problems. A map contains thousands of spectra, so per-pixel signal to noise is traded against acquisition time, and the resulting design question, how short an acquisition still supports the required measurement, is precisely the kind of question the corpus analysed here was collected to answer. It also multiplies the statistical hazards, because adjacent pixels are not independent draws: the sampling step is usually finer than the optical probe, so treating every pixel as a replicate inflates significance. We therefore report within-map significance at an effective sample size rather than at the pixel count, and note that multiple-comparison control [47] is a separate and equally necessary discipline when many bands or many maps are screened at once.

For carbonaceous materials specifically, the fingerprint region is dominated by the D and G bands, conventionally decomposed into five components [48]. The ratio of D to G intensity is the standard disorder metric, but its interpretation depends on excitation wavelength and on whether defects are point-like or line-like [49], so we treat band ratios as protocol-conditional and excitation-conditional quantities throughout rather than as absolute material constants.

3 Related work

Raman software ecosystems. RamanSPy [1, 2] is the closest reference point and supplies the architectural pattern adopted here: typed containers, composable pipelines, named protocols, and an analysis layer that can be benchmarked rather than reimplemented per study. It sits within a broader move toward shared, testable analysis workflows in the sciences [50, 51], and alongside community efforts to standardise Raman practice itself [44, 43]. The ecosystem is predominantly Python [52], built on scikit-learn [53] and, for deep models, on PyTorch [54] or TensorFlow [55].

Chemometrics and machine learning for Raman.

Classical chemometrics, principal components, partial least squares and discriminant analysis, remains the workhorse [34, 35], and automated model search has been used to survey many classifiers cheaply [56]. Deep learning has since been applied widely [57, 58, 59], notably to bacterial and pathogen identification [60, 61, 62], to clinical and diagnostic classification [63, 64, 65, 66, 67], to rapid screening [68, 69, 70], to portable and field instruments [71, 72], and to high-throughput imaging, where a learned model reconstructs usable maps from much shorter exposures [73].

Two observations from this literature shape the present work. First, almost all of it is discriminative: a model

maps a spectrum to a label. Comparatively little of it is *generative* over a map, in the sense of producing a compact object that can regenerate the cube, and that is the gap S3DGS addresses. Second, the same literature repeatedly notes that preprocessing is rarely ablated systematically, so reported performance is entangled with an undeclared chain [42, 43].

Unmixing. Treating a spectrum as a mixture of end-member signatures is standard in remote sensing and transfers directly to Raman imaging. We use Vertex Component Analysis [74], of the same family as N-FINDR [75], with fully constrained least-squares abundances [76]. Spectral angle [77] and information divergence [78] are the conventional similarity measures in that setting [79], and we adopt spectral angle as one of our reconstruction metrics. Physics-constrained autoencoders have recently been proposed for Raman unmixing specifically [80], and offer a learned alternative to the geometric methods used here.

Gaussian splatting and neural fields. Kerbl *et al.* [3] introduced 3D Gaussian Splatting with anisotropic covariances, adaptive density control and a tile-based differentiable rasteriser. Hyperspectral variants followed: HyperGS [4] compresses hyperspectral inputs into a latent space and decodes view-dependent radiance with an MLP, while SpectralGaussians [5] and the diffusion-denoised variant [6] model multi-spectral radiance for scene rendering.

All of these are multi-view rendering methods, and our setting differs in three ways that change the design rather than merely the parameters. There is a single fixed viewpoint, so no projection, occlusion model or view-dependent appearance is required. The third axis is wavenumber rather than depth, so a primitive’s extent along it is a spectral bandwidth and is physically interpretable, which is what makes the fitted objects readable as chemistry. And the data lies on a regular grid, which permits the exact separable evaluation of §7.3 in place of a rasteriser. Implicit neural representations [81] are the natural learned baseline for the same task and are discussed in §12.

4 The ramansp framework

Containers. `Spectrum`, `SpectralImage` and `SpectralVolume` wrap a contiguous intensity array together with a shared monotonic wavenumber axis and a metadata dictionary. The spectral axis is always the last array axis, and is normalised to ascending order on load so that downstream code never has to test for orientation. A `SpectralImage` may be gridded, shape (H, W, K) , or an unstructured bag of spectra, shape (N, K) ; the loader decides which, and refuses to fabricate a grid from coordinates that do not form a clean raster.

Preprocessing. A `Pipeline` is an ordered list of steps drawn from cropping, despiking, baseline correction,

smoothing and normalisation. Named protocols bundle the recipes we use: `carbon_dg` (arPLS baseline [38], then Savitzky and Golay smoothing, then area normalisation), `arpls`, `chord` and `poly3` for straight and polynomial baselines, and `minimal`. The protocol name and its full parameterisation are recorded in the output metadata, so any number computed downstream travels with the recipe that produced it.

One implementation note has a large practical consequence. The arPLS system matrix $\text{diag}(w) + \lambda D^T D$, with D the second-difference operator, is symmetric and pentadiagonal. Solving it with a banded Cholesky factorisation rather than a general sparse solver gives the same matrix and the same answer to 4×10^{-11} relative difference at roughly five times the speed. Whole-corpus reprocessing is what makes a cross-sample study possible at all, and that factor is the difference between a half-hour job and a three-hour one.

Analysis. PCA, NMF and ICA decomposition and k -means clustering wrap scikit-learn. Linear unmixing implements VCA [74] followed by fully-constrained least-squares abundances. Carbon band areas are integrated above local polynomial baselines anchored in windows flanking each band, with a band-free control window integrated by the identical procedure as an acceptance test: a correct baseline returns approximately zero there, so the claim is checkable rather than asserted. Particle and substrate are separated by an Otsu split on log total counts, which requires no noise estimate and therefore avoids the circularity of using a noise threshold to define the mask that defines the noise. Spatial statistics include the variogram, Moran’s I and an AR(1) effective sample size, so that significance within a map can be quoted at the number of independent samples rather than the pixel count.

Corpus construction and anonymisation. The framework walks a study directory and ingests curve-fit tables, matrix-format maps and line scans, single spectra, instrument calibration spectra, and the vendor’s proprietary binaries. It writes a compact, anonymised corpus: one row per acquisition in a manifest, cached arrays for the spectral data, and a machine-readable summary.

A single-pass scanner assigns opaque codes to every specimen identifier, material designation, person name and dated session folder, and the forward map is written to one file under `_private/` and nowhere else. Codes are required to be genuinely opaque: a code built from the initials of the material it replaces still discloses that material to any reader who knows the field, so materials receive sequential labels instead, and the gate described below rejects initial-bearing codes as leaks in their own right. Two names pass through deliberately, because glassy carbon and graphite are public reference standards rather than study identifiers, and naming them is scientifically useful. A separate gate, `run/check_anonymity.py`, scans every shareable output for identifiers and exits non-zero on a

Table 1: Composition of the anonymised corpus.

acquisition kind	count
fit table	40
spectrum	36
raw map	23
autocal	7
raw scan	2
total	108

hit, so the property is enforced rather than intended. On the study analysed here the result is 108 acquisitions (Table 1).

5 Reading the vendor container

Horiba LabSpec6 stores maps and spectra in undocumented binaries (`.16m`, `.16s`). `io.read_16` reads them directly.

Container layout. After an eight-byte LabSpec6 magic and a version word, the file is a flat table of 24-byte records with the fields `[type]` `[id]` `[tag]` `[ptr_hi]` `[val]` `[extra]`. A record whose four-byte tag contains the ASCII `mat` is followed, at record start plus 24 bytes, by a contiguous little-endian `float32` array running to the next record. Several such arrays exist per map: the intensity cube, the wavenumber axis and the two stage-coordinate axes.

The one trap. The `ptr_hi` field looks like a format constant and is not. It is the high half of a serialised 64-bit Windows pointer, so its value (`0x7ff6`, `0x7ff7`, or zero in the 32-bit container variant) records where the writing process happened to be mapped in memory at save time, and it differs between files written by the same software on the same machine. Any parser that hard-codes it works on the file it was developed against and fails silently on the next one. We calibrate it per file from the bytes following a known tag, then use it to walk the record table.

Identification by constraint. The arrays carry no labels, so the reader identifies them by constraint rather than by position. The intensity cube is the largest array. The wavenumber axis is the longest strictly ascending array lying in a plausible Raman-shift range whose length divides the cube. The two stage axes are the *evenly stepped* pair whose lengths satisfy $n_x \cdot n_y \cdot K = |\text{cube}|$; the even-step requirement matters, because intensity data contains ascending runs by chance and an unconstrained search will happily accept one as an axis and reshape the cube wrongly. Which stage axis is the outer one is likewise unlabelled, so both orders are scored by map isotropy: read at the true row length a map is an isotropic two-dimensional field, and read at the wrong one the vertical neighbours

are unrelated points whose differences inflate. If no assignment satisfies the constraint, the reader raises rather than guessing.

Verification. The corpus contains one map for which both the binary and the instrument’s own text export exist, which makes an exact check possible. The recovered cube is bit identical to the export, 766 080 `float32` values with zero differences, and the wavenumber axis, grid dimensions and stage extent all agree. The binary is in fact the more precise of the two, since the text export rounds the axis to six significant figures. Independently of that single pair, across the 57 binaries the reader accepts, the mean spectra place the carbon D and G bands at a median of 1348 and 1580 cm^{-1} respectively, which is where they belong. The first check establishes that the bytes are decoded correctly; the second establishes that the result is physically meaningful across files the first check never saw.

Coverage. The reader handles the container version that carries the experiment maps. It does not yet handle an older 32-bit variant that stores its matrices under a different tag. Those files are overwhelmingly autosave, practice and training scratch rather than experiments, so the shortfall is in bulk rather than in scientific content, but it is a real gap and we state it as one (§11).

6 Cross-sample knowledge graph

6.1 Motivation

A Raman study is not a dataset in the machine-learning sense. It is a heterogeneous accumulation of artefacts: hyperspectral maps, line scans, single cursor spectra, instrument calibration runs, and curve-fit tables exported from the vendor software, produced over months, across several specimens, under several acquisition recipes, by more than one operator. The questions an experimenter actually asks of such a collection are relational. Which acquisitions of different specimens share a spectral signature? Does the disorder metric cluster by material or by acquisition recipe, and if by recipe then the apparent chemistry is an artefact? Are two files the same measurement saved twice? Which map should be reanalysed when a calibration run fails?

None of these is answerable one file at a time, and none is naturally expressed as a table, because the objects are of different kinds and the relations are of different kinds. A property graph is the appropriate structure, and building one also forces the metadata discipline that makes the rest of the analysis trustworthy: an edge cannot be created without deciding what it means and under what preprocessing it was computed.

Table 2: Graph schema. Provenance edges are derived from anonymised paths; content edges are computed from the data and carry the protocol under which they were computed.

Node type	Represents
<code>specimen</code>	a physical sample, opaque code
<code>material</code>	a material or reference class
<code>config</code>	an acquisition configuration
<code>acquisition</code>	one exported artefact
<code>endmember</code>	a VCA component of one map
<code>splatmodel</code>	a fitted Gaussian field
Edge type	Semantics
<code>HAS_ACQUISITION</code>	specimen produced acquisition
<code>OF_MATERIAL</code>	specimen is of material class
<code>USED_CONFIG</code>	acquisition used configuration
<code>HAS_ENDMEMBER</code>	map decomposes into component
<code>MODELLED_BY</code>	map has a fitted field
<code>SIMILAR_TO</code>	mutual k NN in spectral space
<code>SHARES_ENDMEMBER</code>	matched component across specimens
<code>DISORDER_NEIGHBOUR</code>	close on a protocol-stamped metric

6.2 Schema

Nodes and edges are summarised in Table 2. Provenance edges are recovered from the anonymised directory structure. Content edges are computed:

- **SIMILAR_TO**, a mutual k -nearest-neighbour edge with $k = 5$, computed on the mean spectrum resampled to a common axis where spectra exist, and on a z -scored feature vector otherwise. The two bases are never ranked against each other, because they are not on a common scale and mixing them would let a curve-fit table outrank a real spectrum. The tabular basis is flagged `confidence=low`, since such a table carries only two of the five carbon bands and no coordinates.
- **SHARES_ENDMEMBER**, cosine at least 0.9 between VCA endmembers [74] belonging to *different* specimens. This is the edge that expresses a chemical claim rather than a similarity of averages: two specimens may have quite different mean spectra and still contain a common component.
- **DISORDER_NEIGHBOUR**, acquisitions within 0.06 on the bounded disorder index $A_{D1}/(A_{D1} + A_{G+D2})$ and sharing a preprocessing protocol. The protocol condition is not a formality. As §2.2 sets out, a band ratio without its chain is underspecified, so an edge joining two numbers computed under different chains would assert a similarity that the data does not support.

6.3 Why mutual k NN rather than a similarity threshold

An absolute cosine cut is the obvious construction and the wrong one here. Every acquisition in this corpus is carbonaceous and therefore has D and G bands, so spectra

are similar almost by construction. Over all pairs the median cosine is 0.80, with quartiles at 0.55 and 0.92; a cut at 0.65 links two thirds of all pairs. The resulting graph has communities and a respectable modularity, and they describe the threshold rather than the samples.

A mutual k NN rule asks a relative question instead: are these two acquisitions among each other’s nearest neighbours? It is invariant to a monotone rescaling of the similarity, keeps average degree bounded by construction, and degrades gracefully in exactly the regime that defeats a threshold, namely when everything is somewhat similar to everything else. The diagnostic that exposed the problem was simply plotting the distribution of pairwise similarity before choosing a rule rather than after, which we recommend as routine.

6.4 Community structure and interpretation

Communities are extracted by greedy modularity maximisation over the content sub-graph only, so that provenance edges, which are true by construction, cannot manufacture structure. On this corpus the graph has 236 nodes and 600 edges in 23 communities at modularity $Q = 0.68$ (Table 3, Fig. 1), with purity against the sparse material labels of 0.86.

Three results are worth separating from the summary statistics.

Duplicate detection. The graph flags 5 groups of byte-identical curve-fit exports, the same measurement saved repeatedly under different names. This matters beyond tidiness: undetected, such duplicates inflate every count-based statistic over the corpus and, in a similarity-clustered analysis, create artificially tight clusters that invite interpretation as reproducibility.

Connectivity depends on the reader. With text exports alone the graph falls into two disconnected components, a small provenance-linked cluster and a large decontextualised dump of curve-fit tables with no specimen or configuration attached. Reading the vendor binaries (§5) collapses this to essentially one connected object, because the binaries carry the provenance the exports discard. The change is qualitative, not a matter of degree, and it illustrates the argument of §1(ii): what a corpus appears to contain is a property of the reader as much as of the experiment.

What the communities are not. Purity of 0.86 against material labels is computed over a small number of labelled acquisitions and should be read as a description of what the method surfaces on this corpus, not as evidence that these materials separate. We return to this in §11.

6.5 Queries

The graph is written as GraphML and is intended to be queried rather than only plotted. Representative traversals include: all acquisitions of a specimen sharing an endmember with a different specimen; acquisitions whose

Table 3: Knowledge-graph summary.

quantity	value
nodes	236
edges	600
communities	23
modularity Q	0.68
community purity vs. class	0.86

Raman cross-sample knowledge graph (236 nodes, 600 edges, 23 communities, $Q=0.68$)

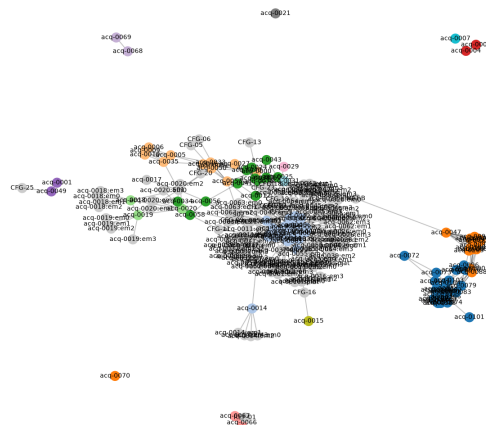


Figure 1: The cross-sample knowledge graph, nodes coloured by community.

disorder index is close under one protocol but not under another, which localises protocol sensitivity to particular samples; maps that have a fitted field and therefore a compact reconstruction available; and configurations whose acquisitions cluster together, which is the signature of an acquisition artefact rather than a chemical one. This last query is the reason `config` is a first-class node: separating chemistry from recipe is only possible if the recipe is in the graph.

7 Spectral 3D Gaussian Splatting

7.1 Representation

Let the map be sampled on a regular grid and rescaled to the unit cube, with position $p = (x, y, z)$ where z is the normalised wavenumber. The field is a sum of N anisotropic three-dimensional Gaussians,

$$\hat{V}(p) = \sum_{i=1}^N a_i \exp\left(-\frac{1}{2}(p - \mu_i)^\top P_i (p - \mu_i)\right), \quad (1)$$

with centre $\mu_i \in \mathbb{R}^3$, amplitude $a_i = e^{\lambda_i} > 0$, and precision

$$P_i = R(q_i) \text{diag}(s_i^{-2}) R(q_i)^\top, \quad s_i = e^{\ell_i} \in \mathbb{R}_{>0}^3, \quad (2)$$

where $R(q_i)$ is the rotation of the unit quaternion q_i . Each primitive carries $3 + 3 + 4 + 1 = 11$ parameters. Ampli-

tude and scale are parameterised logarithmically so that positivity is structural rather than enforced.

Each Gaussian is a genuine ellipsoid in (x, y, ν) . Its two spatial semi-axes localise a chemical domain, and its wavenumber centre and semi-axis localise a band and its width. A carbon particle is therefore reconstructed by ellipsoids clustered over the particle footprint, narrow along z , centred near the D and G positions of the standard assignment [48], with a_i the band intensity. This is what makes the representation legible: a primitive reads directly as “a band, at this position, this strong, this wide”.

We are careful about what that does not mean. The ellipsoids are band *localisers*, not a five-component chemical decomposition. Which defect populations a D/G ratio reflects depends on excitation and defect type [49], and a reconstruction of one cube cannot settle that attribution.

7.2 Objective and analytic gradients

With residual $r(p) = \hat{V}(p) - V(p)$ on the masked voxels Ω ,

$$\mathcal{L} = \frac{1}{2} \sum_{p \in \Omega} r(p)^2 + \lambda_a \sum_i a_i + \lambda_s \sum_i \|\ell_i\|^2. \quad (3)$$

The L_1 term on amplitude drives redundant primitives towards zero so that pruning has something to remove; the term on ℓ discourages degenerate widths.

Writing $d = p - \mu_i$, $\phi_i(p) = \exp(-\frac{1}{2} d^\top P_i d)$ and $w = r \phi_i$, every gradient is closed form:

$$\partial_{\lambda_i} \mathcal{L} = a_i \sum_p w + \lambda_a a_i, \quad \partial_{\mu_i} \mathcal{L} = a_i \sum_p w (P_i d), \quad (4)$$

$$M_i \equiv \partial_{P_i} \mathcal{L} = -\frac{a_i}{2} \sum_p w d d^\top, \quad (5)$$

and, chaining $P_i = R \Lambda R^\top$ with $\Lambda = \text{diag}(s_i^{-2})$,

$$\partial_{\ell_{i,k}} \mathcal{L} = -\frac{2}{s_{i,k}^2} (R^\top M_i R)_{kk} + 2\lambda_s \ell_{i,k}, \quad (6)$$

$$\partial_R \mathcal{L} = 2 M_i R \Lambda, \quad \partial_{q_i} \mathcal{L} = \frac{I - \hat{q}_i \hat{q}_i^\top}{\|\hat{q}_i\|} \sum_k \langle \partial_R \mathcal{L}, \partial_{\hat{q}_{i,k}} R \rangle, \quad (7)$$

the last factor propagating through the normalisation $\hat{q} = q/\|q\|$. Every expression is validated against central finite differences to a relative error below 10^{-4} , as a unit test rather than as a one-off check, so a later optimisation cannot silently break the mathematics.

7.3 Optimisation and complexity

Every fifth wavenumber channel is withheld from Ω and used only for scoring. Quality is reported as PSNR, SSIM [83], spectral angle, and the noise-floor ratio introduced in §8.

Three implementation decisions determine whether this is practical.

Algorithm 1 S3DGS fit (pure NumPy)

```

1: importance-sample  $N$  centres with probability  $\propto |V|$ 
2: initialise  $s$  at a fixed voxel width,  $q$  at identity
3: for  $t = 1, \dots, T$  do
4:    $\hat{V} \leftarrow$  Eq. (1) on each primitive's  $3\sigma$  box
5:    $\mathcal{L}, \nabla \leftarrow$  closed-form gradients (§5.2)
6:   Adam step [82] with per-group rates
7:   clamp  $\ell$  to the permitted width range, in voxels
8:   if  $t \bmod \tau = 0$  and  $t \leq T/2$  then
9:     clone or split primitives with large  $\|\partial_\mu \mathcal{L}\|$ 
10:    prune primitives with  $a_i < \epsilon \text{ median}(a)$ 
11:   end if
12: end for

```

Truncation. Each Gaussian is evaluated only on the grid voxels inside its 3σ box. Cost is therefore $O(N \cdot B)$ per iteration with B the box volume in voxels, independent of the size of the map, which is why a 153×144 map costs about what a 30×32 map costs at the same primitive count. The whole population is advanced as one batched array expression [84] rather than a Python loop over primitives.

Locating the box by search. The box must be centred by searching the axis, not by dividing by a nominal spacing. The fitted wavenumber axis is not uniformly spaced once every fifth channel is withheld, and the drift that results from assuming otherwise is the subject of §9.

Separable evaluation. The quadratic form expands as

$$d^\top P d = \sum_k P_{kk} d_k^2 + 2 \sum_{k < l} P_{kl} d_k d_l, \quad (8)$$

and each offset d_k depends on only one box axis. The $(N, B_y, B_x, B_z, 3)$ offset tensor, which reaches hundreds of megabytes for a few thousand primitives, is therefore never formed, and every gradient contraction reduces to nine scalars per primitive: three first moments for ∂_μ and six second moments for M_i .

Primitive widths are bounded in voxels rather than in absolute units. This keeps the box volume, and hence the cost, constant across maps of very different pixel counts, and it keeps the meaning of a primitive constant too: a primitive should not be much wider than the sampling that produced it.

7.4 Scaling to millions

The NumPy engine is demonstrated at 10^3 primitives and is expected to reach 10^4 to 10^5 on a CPU. The port to the 10^6 to 10^7 regime is the standard tiled rasteriser [3] with a third axis: parameters become CUDA tensors and automatic differentiation replaces §5.2, with the closed-form gradients retained as the test oracle; the volume is partitioned into 16^3 voxel tiles and each primitive is assigned by segmented sort to the tiles its 3σ box intersects, so each tile sums only its own primitives; and densification runs over the full population every hundred steps. An optional extension gives each primitive a length- D code over

Table 4: Fitting hyperparameters. Learning rates are per parameter group, in the convention of [3], where centres must move far more slowly than amplitudes. The primitive count is set from the masked volume so that density rather than count is held approximately fixed across maps.

symbol	value	meaning
T	140	iterations
hold-out	every 5th	withheld channels
bin	4	wavenumber binning
η_μ	4×10^{-4}	centre rate
η_ℓ	6×10^{-3}	log-scale rate
η_q	3×10^{-3}	quaternion rate
η_λ	3×10^{-2}	log-amplitude rate
λ_a	2×10^{-4}	amplitude L_1
λ_s	1×10^{-4}	scale prior
trunc	3σ	evaluation radius
s bounds	0.45 to 2.6 vox	primitive width
τ	40	densify interval
densify until	$T/2$	no growth after
prune	$5 \times 10^{-3}\bar{a}$	amplitude floor

an NMF or VCA endmember dictionary Φ , so that the field value becomes $\sum_i a_i G_i(p) (\Phi^\top c_i)(\nu)$ and every ellipsoid carries a physically realisable Raman profile rather than a scalar amplitude.

8 Results

Setup. All fits run single-threaded on a CPU, in double precision, with no GPU. The flagship map is acq-0042: 72×80 pixels and 292 channels, restricted to its 4172 particle pixels by an Otsu split on log counts. Wavenumber channels are binned by four before fitting, and every fifth surviving channel is withheld.

Flagship reconstruction. S3DGS fits 2912 ellipsoids in 874 seconds. Reconstruction reaches 24.7 dB PSNR on fitted channels and 22.5 dB on withheld channels, with SSIM 0.834 on band maps and a mean spectral angle of 0.611 rad. Because the field is continuous in ν , it stands in for the full-resolution cube: 32032 stored numbers replace 1218224 masked cube values, a $38.0\times$ reduction, or $9.5\times$ counted only against the binned channels the fit actually saw. Band maps rendered from the field reproduce the integrated band-area maps (Fig. 4), and per-pixel spectra including withheld channels are recovered (Fig. 5). The fitted primitives organise themselves into band-centred sheets over the particle footprint (Fig. 3).

Across maps, and what actually limits the fit. Fitting 6 maps spanning different acquisition recipes gives held-out PSNR from 10.9 to 27.2 dB at 8.1 to $38.0\times$ compression (Table 5). Read naively that is a wide and unflattering spread. It is also the wrong reading, and the reason is worth stating plainly: **PSNR against a measured target confounds the representation with the**

Table 5: S3DGS across acquisitions. ρ is primitives per 1000 fitted voxels; PSNR is on wavenumber channels withheld from the fit; r/σ is that residual over the map’s own measurement noise, where 1.0 is the noise floor.

map	N	ρ	compr.	PSNR (dB)	r/σ
map 0042	2912	12	38.0	22.5	1.06
map 0039	2912	13	36.2	22.9	1.02
map 0028	2912	13	34.5	16.9	1.12
map 0059	2912	15	29.6	19.1	1.09
map 0016	2912	24	19.1	10.9	1.43
map 0055	1239	56	8.1	27.2	0.70

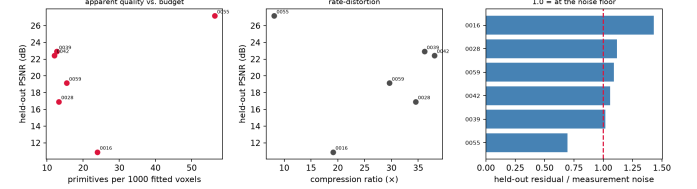


Figure 2: Left: held-out PSNR against primitive density. Middle: the rate-distortion trade-off. Right: held-out residual divided by each map’s own measurement noise. The dashed line at 1.0 is the noise floor, below which no representation can go on that data.

measurement. These acquisitions differ in exposure, and their contrast-to-noise ratios differ by more than a factor of six, from 28 on the noisiest map to 186 on the cleanest.

The honest comparison is the held-out residual against an *independent* estimate of each map’s own noise, obtained from second differences along wavenumber, since a band spans tens of channels and noise does not. That ratio r/σ runs from 0.70 to 1.43 across the set (Fig. 2, right). Five of 6 maps sit within 12% of their own noise floor, and the best sits *below* it, because a smooth field of ellipsoids cannot represent white noise and therefore discards it. The low-PSNR maps are not badly reconstructed. They are noisily measured, and the field is declining to fit the noise.

That interpretation is testable and we test it, on synthetic data where the clean signal is known: a field fitted to a noisy cube ends up closer to the underlying clean signal than the noisy cube it was fitted to. The property is kept as a unit test rather than a figure.

A caveat we do not paper over. The primitive budget is capped, so density ρ is not constant across the set, running from 12 to 56 primitives per 1000 voxels, and the larger maps are the more thinly parameterised. Quality does track ρ (Fig. 2, left), with the noisiest map as the clear outlier, exactly as the noise-floor account predicts. A clean rate-distortion study would sweep ρ on a single map rather than inferring the trend across maps of differing difficulty, and we flag that as the correct next experiment rather than claiming the present one substitutes for it.

2912 ellipsoids (top 900 by opacity)

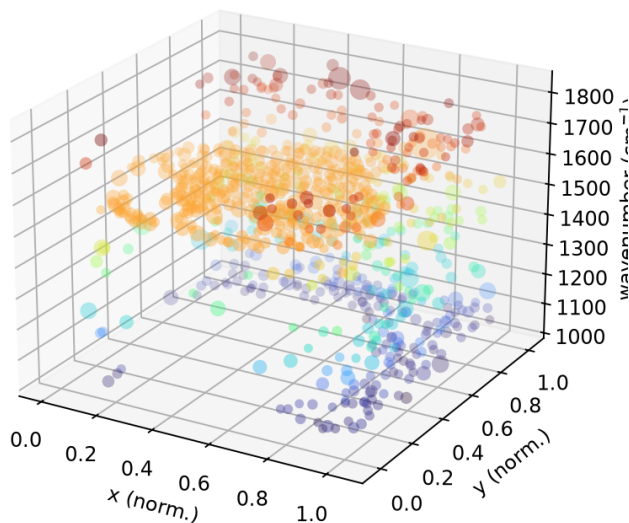


Figure 3: The fitted ellipsoid field, plotted at each primitive’s centre in (x, y, ν) and coloured by wavenumber. The primitives self-organise into band-centred sheets over the particle footprint.

Preprocessing dependence. Fig. 6 recomputes the bounded disorder index on the flagship map under 5 baseline families. Unlike the raw D/G ratio, which prior work on this corpus found to move by a factor of about 1.8 across defensible baselines, the bounded index is stable to 3.1% (0.275 to 0.284). That stability is the reason it, and not a raw ratio, is the quantity carried on graph edges. Stability on one map is not a guarantee, so the protocol is recorded on every `DISORDER_NEIGHBOUR` edge regardless.

9 Failure modes encountered

Three of the defects we hit produced plausible numbers rather than errors. We report them because each is a trap the next implementation will meet, and because the mechanism that caught each one is reusable.

A truncation box that walked off its own Gaussian. The 3σ box was originally located by dividing the offset from the axis origin by a median channel spacing. That is correct for a uniform axis. The fitted wavenumber axis is not uniform: withholding every fifth channel leaves gaps of 1, 1, 1, 2 repeating, so the index estimate drifts by about a quarter of an index per channel and, after sixty channels, the box has left the primitive it belongs to. The field then loses exactly the contributions nearest each centre. Nothing raised, the loss still decreased, and the rendered maps still looked correct. What caught it was compar-

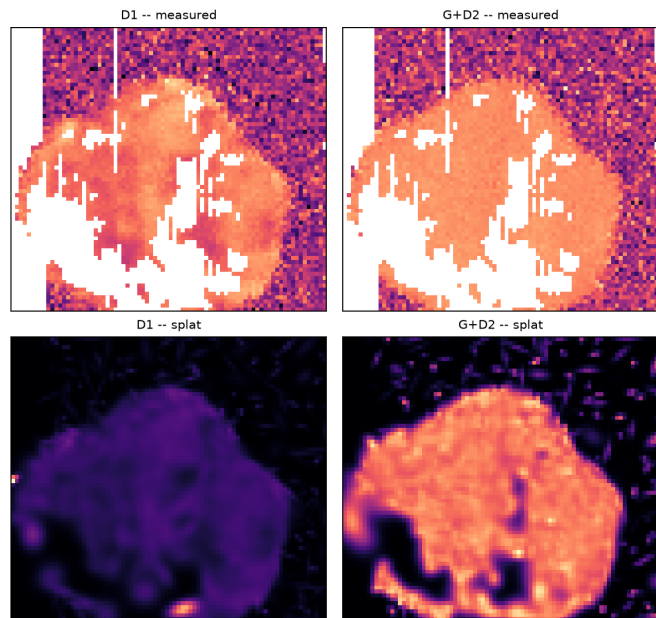


Figure 4: Measured integrated band-area maps (top) against the same bands rendered from the Gaussian field (bottom), for $D1$ and $G+D2$.

ing the truncated evaluator against the exact dense one on the same points, which is now a regression test on a deliberately non-uniform axis.

A similarity threshold that encoded itself. Building `SIMILAR_TO` from an absolute cosine cut produced a graph with communities and a respectable modularity. It was measuring the threshold: at 0.65, two thirds of all pairs are linked, because every spectrum in a carbon corpus has D and G bands. The diagnostic was to plot the distribution of pairwise similarity before choosing a rule, rather than after.

An anonymisation leak from a derived column. The path scrubber worked correctly, and material names still reached the published tables, because a downstream dictionary mapped anonymised codes back to human-readable class labels for plotting. Scrubbing inputs is not sufficient when derived columns can reintroduce what was removed. The gate that now runs over the outputs also caught a false positive of its own, a five-digit specimen pattern matching inside a hexadecimal content digest, which is why its identifier patterns require alphanumeric rather than digit boundaries.

10 Discussion

What the reader changes, and why it is a scientific rather than an engineering point. It is tempting to

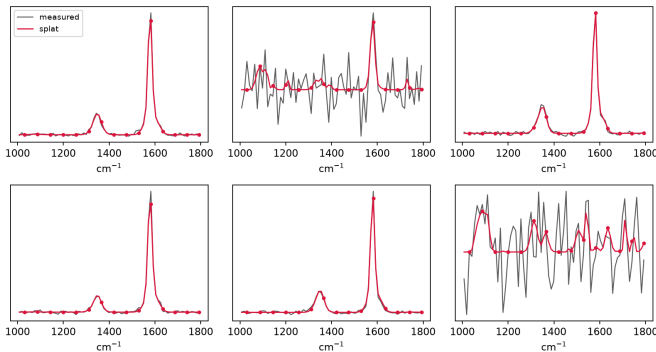


Figure 5: Measured (grey) against reconstructed (red) spectra at random particle pixels. Dots mark the withheld wavenumber channels, which the field never saw.

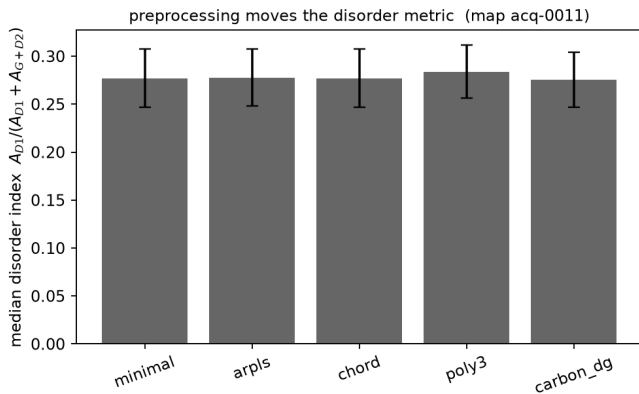


Figure 6: Median bounded disorder index on the flagship map across 5 preprocessing protocols. Error bars are half the interquartile range.

file a binary-format reader under infrastructure. The effect here is on what could be concluded. With text exports alone the corpus held one hyperspectral map, the knowledge graph was disconnected, and most curve-fit tables carried no provenance at all. With the binaries read directly it holds 23 maps and the graph is essentially connected. Nothing about the experiment changed; the experiment had always produced this data. What changed is the selection mechanism between the instrument and the analysis, which was previously an operator deciding which files to export by hand. Any conclusion drawn from the exported subset was conditioned on that decision, and the conditioning was invisible. Community efforts to standardise Raman practice have concentrated, rightly, on acquisition and preprocessing [44, 43]; the path from instrument storage to analysis deserves the same scrutiny.

Metrics that measure the instrument rather than the method. The clearest methodological result here is negative. Held-out PSNR across our maps spans more

than 16 dB, which reads as a representation that works on some data and fails on others. Referred to each map’s own measurement noise, the same fits sit between 0.70 and 1.43 of the noise floor, which reads as a representation whose error is set by the acquisitions. The second description is the correct one, and the difference is not a matter of presentation: the first would lead an experimenter to tune the model, and the second to lengthen the exposure. We suspect this confound is common wherever reconstruction quality is reported on measured rather than synthetic data, and the remedy is cheap, since an independent noise estimate is available from the spectra themselves.

Interpretability of the primitives. Unlike a latent code or a network weight, a fitted ellipsoid has units. Its wavenumber centre is a band position, its extent along that axis a bandwidth, its amplitude an intensity, and its spatial semi-axes a domain size. This places S3DGS closer to physically constrained decompositions [80] than to generic compressors, and it means the representation can be inspected for plausibility rather than only scored. We are deliberate about the limit of that claim: the primitives localise bands, they do not assign them to chemical species, and the five-component carbon model [48] is not identifiable from these spectra in any case.

Where the framework sits relative to learned models. The dominant use of machine learning in Raman analysis is discriminative, mapping a spectrum to a class [57, 58, 60, 63]. S3DGS is generative over a single map and learns nothing transferable between maps: each field is fitted independently, with no training set and no risk of leakage between specimens. That is a limitation when many similar maps are available, and an advantage when they are not, which is the common case in a materials-characterisation study with a handful of specimens. It also composes with the discriminative literature rather than competing with it, since a fitted field supplies denoised spectra and compact per-pixel descriptors that a classifier can consume. Approaches that shorten acquisition by learning to reconstruct [73] address the same underlying economics from the opposite direction.

11 Limitations

Scale. The CPU engine is demonstrated at 10^3 primitives. §7.4 specifies, but does not implement, the GPU path to millions, and we make no measured claim about that regime.

Vendor coverage. 563 files remain unread, in an older 32-bit container variant that stores matrices under a different tag. They are almost entirely autosave, practice and training scratch, so the scientific loss is small, but the number is large and we quote it rather than the flattering one.

Identifiability. Prior work on this corpus established that the instrument’s five-band carbon fit is not identifiable from these spectra: opening the bounds does not improve the answer so much as change the model. S3DGS reconstructs the *cube*, and its primitives are band localisers. Nothing here should be read as a claim that five specific chemical components were resolved.

Statistical power. The corpus contains 3 distinguishable specimen codes and 16 acquisition configurations, and 40 curve-fit tables carry no provenance link at all. Community structure and purity are therefore illustrative of what the method surfaces, not a population-level result about these materials.

Scope of the noise-floor claim. The claim that reconstruction reaches the measurement noise floor is specific to noise-limited data. On a nearly noise-free map the residual is dominated by representation error and the ratio exceeds one, correctly. The ratio is a diagnostic of what limits a given fit, not a score to be maximised.

12 Future work

A GPU backend, and what it would buy. The tiled rasteriser of §7.4 is specified and not implemented. Beyond speed, the 10^6 regime changes what the representation can express: at present a primitive is constrained to be a few voxels wide so that cost stays bounded, and with a GPU that constraint could be relaxed to let large smooth regions be captured by few wide primitives and detailed regions by many narrow ones, which is the adaptive behaviour that makes 3DGS effective in vision [3]. The closed-form gradients derived here become the test oracle for the autodiff implementation rather than being discarded.

Spectral dictionaries and physical priors. Giving each primitive a code over an endmember dictionary, so that the field value is $\sum_i a_i G_i(p)(\Phi^\top c_i)(\nu)$, would make every ellipsoid carry a realisable Raman profile rather than a scalar amplitude. The dictionary could come from VCA [74], from a physics-constrained autoencoder [80], or from a reference database such as RRUFF [27]. A stronger variant replaces the isotropic spectral extent with an explicit line-shape model, Lorentzian for G and Gaussian for the disorder bands as the instrument’s own fitting software assumes, which would make band width a fitted physical parameter rather than an emergent one.

A proper rate-distortion study. As noted in §8, our compression figures vary across maps that also vary in difficulty. The correct experiment sweeps primitive count on a single map, holding the data fixed, to produce a genuine rate-distortion curve, and repeats it at several noise levels

to show where the curve saturates at the noise floor. This is inexpensive and is the first thing we would add.

Neural-field baselines and hybrid models. A Fourier-featured multilayer perceptron mapping (x, y, ν) to intensity [81] is the natural learned counterpart and would establish whether explicit primitives buy interpretability at a cost in fidelity, or at no cost. A hybrid in which a coarse field is explicit and a small network models the residual is the obvious third option.

Extending the graph. Three extensions follow directly from the schema. Learned node embeddings would let similarity be defined over provenance and content jointly rather than content alone. Temporal edges would expose drift, since the corpus spans months and includes the calibration runs needed to detect it, connecting to the instrument-stability concerns raised in interlaboratory work [31]. And treating each acquisition configuration as a design point would turn the graph into the analysis substrate for the experimental design the corpus was collected to serve, with response variables computed from cubes rather than scored by eye.

Beyond carbon. Nothing in the framework is specific to carbonaceous materials except the default band windows and the disorder index. The reader, the graph and the splatting representation apply unchanged to any Raman mapping study, and the biological and clinical imaging settings [45, 46, 9] are where map counts, and therefore the value of a cross-sample view, are largest.

13 Reproducibility

The framework installs as a package and its test suite covers container round-trips, pipeline determinism, the analytic gradients against finite differences, the truncated evaluator against the dense one, graph construction, anonymisation, and the denoising property. The five pipeline stages run end to end from the raw study directory, and every number quoted in this manuscript is emitted by the final stage into a macro file from the pipeline’s own outputs, so that no figure in the text can drift from the data that produced it. Reconstruction metrics can be recomputed from stored fields without refitting, which is what made adding the noise-floor analysis after the fits a matter of seconds rather than hours.

14 Conclusion

We set out to make a months-long Raman mapping study analysable as a whole rather than a file at a time, and three things were needed.

The first was to stop treating the vendor’s format as a boundary. Reading the LabSpec6 container directly, verified bit identical against the instrument’s own export,

took the analysable corpus from one hyperspectral map to 23 and turned a disconnected graph into a connected one. The subset an operator happens to export is a selection mechanism sitting between the experiment and its conclusions, and it is worth removing rather than documenting.

The second was to give the corpus a relational structure. The knowledge graph makes cross-sample questions expressible, detects duplicate exports that would otherwise inflate every count, and enforces at the level of data structure the discipline that §2.2 argues for: a band ratio travels with the preprocessing protocol that produced it, and ratios computed under different protocols are never compared. Building it also required rejecting the obvious similarity rule, because in a corpus where every spectrum has D and G bands an absolute threshold measures itself.

The third was Spectral 3D Gaussian Splatting, which gives each map a compact, differentiable and interpretable representation adapted from scene reconstruction to a setting with one viewpoint and a physically meaningful third axis. Each primitive has units and reads as a band at a position with a width and an intensity. Across 6 maps and acquisition recipes its held-out error lies between 0.70 and 1.43 of each map’s own measurement noise, at 8.1 to 38.0× compression. That is the result we would emphasise, and it is one a PSNR table alone would have hidden: the reconstruction is limited by the acquisitions, not by the model, and on the cleanest map the field is below the noise floor because it declines to encode noise.

We have also reported the three defects that produced plausible numbers rather than errors, since each cost real time and each will be met again by anyone building the same thing. The next steps are the GPU backend of §7.4, spectral dictionaries that give primitives physical line shapes, a single-map rate-distortion sweep, and a neural-field baseline [81].

Code and data availability. The RAMANSP package, the pipeline scripts, the anonymity gate and this manuscript’s asset generator are released together. The corpus is distributed in anonymised form only, and the identity map is withheld.

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